

4 (**short**) The specific work output w produced when a perfect gas flows steadily through a given turbine depends only on the gas inlet velocity V_1 , the gas inlet temperature T_1 , the specific heat capacity at constant pressure c_p , the specific gas constant R , and the blade speed U within the turbine.

(a) What is the minimum number of non-dimensional groups according to Buckingham's rule? [4]

(b) Suggest a possible complete set of non-dimensional groups. [6]